

Random Variables and Their Distributions

19 problems

11 textbook exercises · 3 strategic practice · 5 homework

Questions below; full solutions at the end of the chapter.

TEXTBOOK EXERCISES

PMFs and CDFs

Q1 Book Ch.3, Exercise 6

[solution · p. 77](#)

6. (S) *Benford's law* states that in a very large variety of real-life data sets, the first digit approximately follows a particular distribution with about a 30% chance of a 1, an 18% chance of a 2, and in general

$$P(D = j) = \log_{10} \left(\frac{j+1}{j} \right), \text{ for } j \in \{1, 2, 3, \dots, 9\},$$

where D is the first digit of a randomly chosen element. Check that this is a valid PMF (using properties of logs, not with a calculator).

Q2 Book Ch.3, Exercise 11

[solution · p. 77](#)

11. (S) Let X be an r.v. whose possible values are $0, 1, 2, \dots$, with CDF F . In some countries, rather than using a CDF, the convention is to use the function G defined by $G(x) = P(X < x)$ to specify a distribution. Find a way to convert from F to G , i.e., if F is a known function, show how to obtain $G(x)$ for all real x .

Named distributions

Q3 Book Ch.3, Exercise 18

[solution · p. 77](#)

18. (S) (a) In the World Series of baseball, two teams (call them A and B) play a sequence of games against each other, and the first team to win four games wins the series. Let

p be the probability that A wins an individual game, and assume that the games are independent. What is the probability that team A wins the series?

(b) Give a clear intuitive explanation of whether the answer to (a) depends on whether the teams always play 7 games (and whoever wins the majority wins the series), or the teams stop playing more games as soon as one team has won 4 games (as is actually the case in practice: once the match is decided, the two teams do not keep playing more games).

Q4 Book Ch.3, Exercise 21

[solution · p. 78](#)

21. (S) Let $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$, independent of X . Show that $X - Y$ is *not* Binomial.

Q5 Book Ch.3, Exercise 25

[solution · p. 78](#)

25. (S) Alice flips a fair coin n times and Bob flips another fair coin $n + 1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n + 1, \frac{1}{2})$.

(a) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.

(b) Compute $P(X < Y)$.

Hint: Use (a) and the fact that X and Y are integer-valued.

Q6 Book Ch.3, Exercise 28

[solution · p. 79](#)

28. (S) There are n eggs, each of which hatches a chick with probability p (independently). Each of these chicks survives with probability r , independently. What is the distribution of the number of chicks that hatch? What is the distribution of the number of chicks that survive? (Give the PMFs; also give the names of the distributions and their parameters, if applicable.)

Q7 Book Ch.3, Exercise 29

[solution · p. 79](#)

29. (S) A sequence of n independent experiments is performed. Each experiment is a success with probability p and a failure with probability $q = 1 - p$. Show that conditional on the number of successes, all valid possibilities for the list of outcomes of the experiment are equally likely.

Q8 Book Ch.3, Exercise 35

[solution · p. 79](#)

35. (S) Players A and B take turns in answering trivia questions, starting with player A answering the first question. Each time A answers a question, she has probability p_1 of getting it right. Each time B plays, he has probability p_2 of getting it right.

(a) If A answers m questions, what is the PMF of the number of questions she gets right?

(b) If A answers m times and B answers n times, what is the PMF of the total number of questions they get right (you can leave your answer as a sum)? Describe exactly when/whether this is a Binomial distribution.

(c) Suppose that the first player to answer correctly wins the game (with no predetermined maximum number of questions that can be asked). Find the probability that A wins the game.

37. (S) A message is sent over a noisy channel. The message is a sequence $x_1, x_2, \dots, x_n$ of n bits ($x_i \in \{0, 1\}$). Since the channel is noisy, there is a chance that any bit might be corrupted, resulting in an error (a 0 becomes a 1 or vice versa). Assume that the error events are independent. Let p be the probability that an individual bit has an error ($0 < p < 1/2$). Let $y_1, y_2, \dots, y_n$ be the received message (so $y_i = x_i$ if there is no error in that bit, but $y_i = 1 - x_i$ if there is an error there).

To help detect errors, the n th bit is reserved for a parity check: x_n is defined to be 0 if $x_1 + x_2 + \dots + x_{n-1}$ is even, and 1 if $x_1 + x_2 + \dots + x_{n-1}$ is odd. When the message is received, the recipient checks whether y_n has the same parity as $y_1 + y_2 + \dots + y_{n-1}$. If the parity is wrong, the recipient knows that at least one error occurred; otherwise, the recipient assumes that there were no errors.

(a) For $n = 5, p = 0.1$, what is the probability that the received message has errors which go undetected?

(b) For general n and p , write down an expression (as a sum) for the probability that the received message has errors which go undetected.

(c) Give a simplified expression, not involving a sum of a large number of terms, for the probability that the received message has errors which go undetected.

Hint for (c): Letting

$$a = \sum_{k \text{ even}, k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} \text{ and } b = \sum_{k \text{ odd}, k \geq 1} \binom{n}{k} p^k (1-p)^{n-k},$$

the binomial theorem makes it possible to find simple expressions for $a + b$ and $a - b$, which then makes it possible to obtain a and b .

Independence of r.v.s

Q10 Book Ch.3, Exercise 42

solution · p. 81

42. (S) Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, etc. (so X takes values $1, 2, \dots, 7$, with equal probabilities). Let Y be the next day after X (again represented as an integer between 1 and 7). Do X and Y have the same distribution? What is $P(X < Y)$?

Mixed practice

Q11 Book Ch.3, Exercise 45

solution · p. 82

45. (S) A new treatment for a disease is being tested, to see whether it is better than the standard treatment. The existing treatment is effective on 50% of patients. It is believed initially that there is a $2/3$ chance that the new treatment is effective on 60% of patients, and a $1/3$ chance that the new treatment is effective on 50% of patients. In a pilot study, the new treatment is given to 20 random patients, and is effective for 15 of them.

(a) Given this information, what is the probability that the new treatment is better than the standard treatment?

(b) A second study is done later, giving the new treatment to 20 new random patients. Given the results of the first study, what is the PMF for how many of the new patients the new treatment is effective on? (Letting p be the answer to (a), your answer can be left in terms of p .)

STRATEGIC PRACTICE

4 Bernoulli and Binomial

Q12 SP 3 §4 · Problem 1

[solution · p. 82](#)

1. (a) In the World Series of baseball, two teams (call them A and B) play a sequence of games against each other, and the first team to win four games wins the series. Let p be the probability that A wins an individual game, and assume that the games are independent. What is the probability that team A wins the series?

(b) Give a clear intuitive explanation of whether the answer to (a) depends on whether the teams always play 7 games (and whoever wins the majority wins the series), or the teams stop playing more games as soon as one team has won 4 games (as is actually the case in practice: once the match is decided, the two teams do not keep playing more games).

Q13 SP 3 §4 · Problem 2

[solution · p. 84](#)

2. A sequence of n independent experiments is performed. Each experiment is a success with probability p and a failure with probability $q = 1 - p$. Show that conditional on the number of successes, all possibilities for the list of outcomes of the experiment are equally likely (of course, we only consider lists of outcomes where the number of successes is consistent with the information being conditioned on).

Q14 SP 3 §4 · Problem 3

[solution · p. 85](#)

3. Let $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$, independent of X .
 - (a) Show that $X + Y \sim \text{Bin}(n + m, p)$, using a story proof.
 - (b) Show that $X - Y$ is *not* Binomial.
 - (c) Find $P(X = k | X + Y = j)$. How does this relate to the elk problem from HW 1?

HOMEWORK

Q15 HW 3 · Problem 6

[solution · p. 86](#)

6. Players A and B take turns in answering trivia questions, starting with player A answering the first question. Each time A answers a question, she has probability p_1 of getting it right. Each time B plays, he has probability p_2 of getting it right.

(a) If A answers m questions, what is the PMF of the number of questions she gets right?

(b) If A answers m times and B answers n times, what is the PMF of the total number of questions they get right (you can leave your answer as a sum)? Describe exactly when/whether this is a Binomial distribution.

(c) Suppose that the first player to answer correctly wins the game (with no predetermined maximum number of questions that can be asked). Find the probability that A wins the game.

Q16 HW 3 · Problem 7

solution · p. 87

7. A message is sent over a noisy channel. The message is a sequence $x_1, x_2, \dots, x_n$ of n bits ($x_i \in \{0, 1\}$). Since the channel is noisy, there is a chance that any bit might be corrupted, resulting in an error (a 0 becomes a 1 or vice versa). Assume that the error events are independent. Let p be the probability that an individual bit has an error ($0 < p < 1/2$). Let $y_1, y_2, \dots, y_n$ be the received message (so $y_i = x_i$ if there is no error in that bit, but $y_i = 1 - x_i$ if there is an error there).

To help detect errors, the n th bit is reserved for a parity check: x_n is defined to be 0 if $x_1 + x_2 + \dots + x_{n-1}$ is even, and 1 if $x_1 + x_2 + \dots + x_{n-1}$ is odd. When the message is received, the recipient checks whether y_n has the same parity as $y_1 + y_2 + \dots + y_{n-1}$. If the parity is wrong, the recipient knows that at least one error occurred; otherwise, the recipient assumes that there were no errors.

(a) For $n = 5, p = 0.1$, what is the probability that the received message has errors which go undetected?

(b) For general n and p , write down an expression (as a sum) for the probability that the received message has errors which go undetected.

(c) Give a simplified expression, not involving a sum of a large number of terms, for the probability that the received message has errors which go undetected.

Hint for (c): Letting

$$a = \sum_{k \text{ even}, k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} \text{ and } b = \sum_{k \text{ odd}, k \geq 1} \binom{n}{k} p^k (1-p)^{n-k},$$

the binomial theorem makes it possible to find simple expressions for $a + b$ and $a - b$, which then makes it possible to obtain a and b .

1. Let X be a r.v. whose possible values are $0, 1, 2, \dots$, with CDF F . In some countries, rather than using a CDF, the convention is to use the function G defined by $G(x) = P(X < x)$ to specify a distribution. Find a way to convert from F to G , i.e., if F is a known function show how to obtain $G(x)$ for all real x .

2. There are n eggs, each of which hatches a chick with probability p (independently). Each of these chicks survives with probability r , independently. What is the distribution of the number of chicks that hatch? What is the distribution of the number of chicks that survive? (Give the PMFs; also give the names of the distributions and their parameters, if they are distributions we have seen in class.)

5. A scientist wishes to study whether men or women are more likely to have a certain disease, or whether they are equally likely. A random sample of m women and n men is gathered, and each person is tested for the disease (assume for this problem that the test is completely accurate). The numbers of women and men in the sample who have the disease are X and Y respectively, with $X \sim \text{Bin}(m, p_1)$ and $Y \sim \text{Bin}(n, p_2)$. Here p_1 and p_2 are unknown, and we are interested in testing the “null hypothesis” $p_1 = p_2$.

(a) Consider a 2 by 2 table listing with rows corresponding to disease status and columns corresponding to gender, with each entry the count of how many people have that disease status and gender (so $m + n$ is the sum of all 4 entries). Suppose that it is observed that $X + Y = r$.

The *Fisher exact test* is based on conditioning on both the row and column sums, so m, n, r are all treated as fixed, and then seeing if the observed value of X is “extreme” compared to this conditional distribution. Assuming the null hypothesis, use Bayes’ Rule to find the conditional PMF of X given $X + Y = r$. Is this a distribution we have studied in class? If so, say which (and give its parameters).

(b) Give an intuitive explanation for the distribution of (a), explaining how this problem relates to other problems we’ve seen, and why p_1 disappears (magically?) in the distribution found in (a).

CHAPTER 3 · SOLUTIONS

S1 Book Ch.3, Exercise 6

[question · p. 71](#)

Solution: The function $P(D = j)$ is nonnegative and the sum over all values is

$$\sum_{j=1}^9 \log_{10} \frac{j+1}{j} = \sum_{j=1}^9 (\log_{10}(j+1) - \log_{10}(j)).$$

All terms cancel except $\log_{10} 10 - \log_{10} 1 = 1$ (this is a *telescoping series*). Since the values add to 1 and are nonnegative, $P(D = j)$ is a PMF.

S2 Book Ch.3, Exercise 11

[question · p. 71](#)

Solution: Write

$$G(x) = P(X \leq x) - P(X = x) = F(x) - P(X = x).$$

If x is not a nonnegative integer, then $P(X = x) = 0$ so $G(x) = F(x)$. For x a nonnegative integer,

$$P(X = x) = F(x) - F(x - 1/2)$$

since the PMF corresponds to the lengths of the jumps in the CDF. (The $1/2$ was chosen for concreteness; we also have $F(x - 1/2) = F(x - a)$ for any $a \in (0, 1]$.) Thus,

$$G(x) = \begin{cases} F(x) & \text{if } x \notin \{0, 1, 2, \dots\} \\ F(x - 1/2) & \text{if } x \in \{0, 1, 2, \dots\}. \end{cases}$$

More compactly, we can also write $G(x) = \lim_{t \rightarrow x^-} F(t)$, where the $-$ denotes taking a limit from the left (recall that F is right continuous), and $G(x) = F(\lceil x \rceil - 1)$, where $\lceil x \rceil$ is the ceiling of x (the smallest integer greater than or equal to x).

S3 Book Ch.3, Exercise 18

[question · p. 71](#)

Solution:

(a) Let $q = 1 - p$. First let us do a direct calculation:

$$\begin{aligned} P(\text{A wins}) &= P(\text{A wins in 4 games}) + P(\text{A wins in 5 games}) \\ &\quad + P(\text{A wins in 6 games}) + P(\text{A wins in 7 games}) \\ &= p^4 + \binom{4}{3} p^4 q + \binom{5}{3} p^4 q^2 + \binom{6}{3} p^4 q^3. \end{aligned}$$

To understand how these probabilities are calculated, note for example that

$$\begin{aligned} P(\text{A wins in 5}) &= P(\text{A wins 3 out of first 4}) \cdot P(\text{A wins 5th game} | \text{A wins 3 out of first 4}) \\ &= \binom{4}{3} p^3 q p. \end{aligned}$$

(Each of the 4 terms in the expression for $P(\text{A wins})$ can also be found using the PMF of a distribution known as the *Negative Binomial*, which is introduced in Chapter 4.)

An neater solution is to use the fact (explained in the solution to Part (b)) that we can assume that the teams play all 7 games no matter what. Let X be the number of wins for team A, so that $X \sim \text{Bin}(7, p)$. Then

$$\begin{aligned} P(X \geq 4) &= P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) \\ &= \binom{7}{4} p^4 q^3 + \binom{7}{5} p^5 q^2 + \binom{7}{6} p^6 q + p^7, \end{aligned}$$

which looks different from the above but is actually identical as a function of p (as can be verified by simplifying both expressions as polynomials in p).

(b) The answer to (a) does not depend on whether the teams play all seven games no matter what. Imagine telling the players to continue playing the games even after the match has been decided, just for fun: the outcome of the match won't be affected by this, and this also means that the probability that A wins the match won't be affected by assuming that the teams always play 7 games!

S4 Book Ch.3, Exercise 21

[question · p. 72](#)

Solution: A Binomial can't be negative, but $X - Y$ is negative with positive probability.

S5 Book Ch.3, Exercise 25

[question · p. 72](#)

Solution:

(a) Note that $n - X \sim \text{Bin}(n, 1/2)$ and $n + 1 - Y \sim \text{Bin}(n + 1, 1/2)$ (we can interpret this by thinking of counting Tails rather than counting Heads), with $n - X$ and $n + 1 - Y$ independent. So $P(X < Y) = P(n - X < n + 1 - Y)$, since both sides have exactly the same structure.

(b) We have

$$P(X < Y) = P(n - X < n + 1 - Y) = P(Y < X + 1) = P(Y \leq X)$$

since X and Y are integer-valued (e.g., $Y < 5$ is equivalent to $Y \leq 4$). But $Y \leq X$ is the complement of $X < Y$, so $P(X < Y) = 1 - P(X < Y)$. Thus, $P(X < Y) = 1/2$.

Solution:



Let H be the number of eggs that hatch and X be the number of hatchlings that survive. Think of each egg as a Bernoulli trial, where for H we define “success” to mean hatching, while for X we define “success” to mean surviving. For example, in the picture above, where denotes an egg that hatches with the chick surviving, denotes an egg that hatched but whose chick died, and denotes an egg that didn't hatch, the events $H = 7$, $X = 5$ occurred. By the story of the Binomial, $H \sim \text{Bin}(n, p)$, with PMF

$$P(H = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

for $k = 0, 1, \dots, n$.

The eggs independently have probability pr each of hatching a chick that survives. By the story of the Binomial, we have $X \sim \text{Bin}(n, pr)$, with PMF

$$P(X = k) = \binom{n}{k} (pr)^k (1 - pr)^{n-k}$$

for $k = 0, 1, \dots, n$.

Solution:

Let X_j be 1 if the j th experiment is a success and 0 otherwise, and let $X = X_1 + \dots + X_n$ be the total number of successes. Then for any k and any $a_1, \dots, a_n \in \{0, 1\}$ with $a_1 + \dots + a_n = k$,

$$\begin{aligned} P(X_1 = a_1, \dots, X_n = a_n | X = k) &= \frac{P(X_1 = a_1, \dots, X_n = a_n, X = k)}{P(X = k)} \\ &= \frac{P(X_1 = a_1, \dots, X_n = a_n)}{P(X = k)} \\ &= \frac{p^k q^{n-k}}{\binom{n}{k} p^k q^{n-k}} \\ &= \frac{1}{\binom{n}{k}}. \end{aligned}$$

This does not depend on $a_1, \dots, a_n$. Thus, for n independent Bernoulli trials, given that there are exactly k successes, the $\binom{n}{k}$ possible sequences consisting of k successes and $n - k$ failures are equally likely. Interestingly, the conditional probability above also does not depend on p (this is closely related to the notion of a *sufficient statistic*, which is an important concept in statistical inference).

Solution:

(a) The r.v. is $\text{Bin}(m, p_1)$, so the PMF is $\binom{m}{k} p_1^k (1 - p_1)^{m-k}$ for $k \in \{0, 1, \dots, m\}$.

(b) Let T be the total number of questions they get right. To get a total of k questions right, it must be that A got 0 and B got k , or A got 1 and B got $k - 1$, etc. These are disjoint events so the PMF is

$$P(T = k) = \sum_{j=0}^k \binom{m}{j} p_1^j (1 - p_1)^{m-j} \binom{n}{k-j} p_2^{k-j} (1 - p_2)^{n-(k-j)}$$

for $k \in \{0, 1, \dots, m + n\}$, with the usual convention that $\binom{n}{k}$ is 0 for $k > n$.

This is the $\text{Bin}(m + n, p)$ distribution if $p_1 = p_2 = p$ (using the story for the Binomial, or using Vandermonde's identity). For $p_1 \neq p_2$, it's not a Binomial distribution, since the trials have different probabilities of success; having some trials with one probability of success and other trials with another probability of success isn't equivalent to having trials with some "effective" probability of success.

(c) Let $r = P(\text{A wins})$. Conditioning on the results of the first question for each player, we have

$$r = p_1 + (1 - p_1)p_2 \cdot 0 + (1 - p_1)(1 - p_2)r,$$

which gives $r = \frac{p_1}{1 - (1 - p_1)(1 - p_2)} = \frac{p_1}{p_1 + p_2 - p_1 p_2}$.

Solution:

(a) Note that $\sum_{i=1}^n x_i$ is even. If the number of errors is even (and nonzero), the errors will go undetected; otherwise, $\sum_{i=1}^n y_i$ will be odd, so the errors will be detected.

The number of errors is $\text{Bin}(n, p)$, so the probability of undetected errors when $n = 5, p = 0.1$ is

$$\binom{5}{2} p^2 (1-p)^3 + \binom{5}{4} p^4 (1-p) \approx 0.073.$$

(b) By the same reasoning as in (a), the probability of undetected errors is

$$\sum_{k \text{ even}, k \geq 2} \binom{n}{k} p^k (1-p)^{n-k}.$$

(c) Let a, b be as in the hint. Then

$$a + b = \sum_{k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} = 1,$$

$$a - b = \sum_{k \geq 0} \binom{n}{k} (-p)^k (1-p)^{n-k} = (1-2p)^n.$$

Solving for a and b gives

$$a = \frac{1 + (1-2p)^n}{2} \text{ and } b = \frac{1 - (1-2p)^n}{2}.$$

$$\sum_{k \text{ even}, k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} = \frac{1 + (1-2p)^n}{2}.$$

Subtracting off the possibility of no errors, we have

$$\sum_{k \text{ even}, k \geq 2} \binom{n}{k} p^k (1-p)^{n-k} = \frac{1 + (1-2p)^n}{2} - (1-p)^n.$$

Sanity check: Note that letting $n = 5, p = 0.1$ here gives 0.073, which agrees with (a); letting $p = 0$ gives 0, as it should; and letting $p = 1$ gives 0 for n odd and 1 for n even, which again makes sense.

Solution: Yes, X and Y have the same distribution, since Y is also equally likely to represent any day of the week. However, X is likely to be less than Y . Specifically,

$$P(X < Y) = P(X \neq 7) = \frac{6}{7}.$$

In general, if Z and W are *independent* r.v.s with the same distribution, then $P(Z < W) = P(W < Z)$ by symmetry. Here though, X and Y are *dependent*, and we have $P(X < Y) = 6/7, P(X = Y) = 0, P(Y < X) = 1/7$.

Solution:

(a) Let B be the event that the new treatment is better than the standard treatment and let X be the number of people in the study for whom the new treatment is effective. By Bayes' rule and LOTP,

$$\begin{aligned} P(B|X = 15) &= \frac{P(X = 15|B)P(B)}{P(X = 15|B)P(B) + P(X = 15|B^c)P(B^c)} \\ &= \frac{\binom{20}{15}(0.6)^{15}(0.4)^5(\frac{2}{3})}{\binom{20}{15}(0.6)^{15}(0.4)^5(\frac{2}{3}) + \binom{20}{15}(0.5)^{20}(\frac{1}{3})}. \end{aligned}$$

(b) Let Y be how many of the new patients the new treatment is effective for and $p = P(B|X = 15)$ be the answer from (a). Then for $k \in \{0, 1, \dots, 20\}$,

$$\begin{aligned} P(Y = k|X = 15) &= P(Y = k|X = 15, B)P(B|X = 15) + P(Y = k|X = 15, B^c)P(B^c|X = 15) \\ &= P(Y = k|B)P(B|X = 15) + P(Y = k|B^c)P(B^c|X = 15) \\ &= \binom{20}{k}(0.6)^k(0.4)^{20-k}p + \binom{20}{k}(0.5)^{20}(1-p). \end{aligned}$$

(This distribution is *not* Binomial. As in the coin with a random bias problem, the individual outcomes are conditionally independent but not independent. Given the true probability of effectiveness of the new treatment, the pilot study is irrelevant and the distribution is Binomial, but without knowing that, we have a mixture of two different Binomial distributions.)

1. (a) In the World Series of baseball, two teams (call them A and B) play a sequence of games against each other, and the first team to win four games wins the series. Let p be the probability that A wins an individual game, and assume that the games are independent. What is the probability that team A wins the series?

Let $q = 1 - p$. First let us do a direct calculation:

$$\begin{aligned} P(A \text{ wins}) &= P(A \text{ winning in 4 games}) + P(A \text{ winning in 5 games}) \\ &\quad + P(A \text{ wins in 6 games}) + P(A \text{ winning in 7 games}) \\ &= p^4 + \binom{4}{3} p^4 q + \binom{5}{3} p^4 q^2 + \binom{6}{3} p^4 q^3. \end{aligned}$$

To understand how these probabilities are calculated, note for example that

$$\begin{aligned} P(A \text{ wins in 5}) &= P(A \text{ wins 3 out of first 4}) \cdot P(A \text{ wins 5th game} | A \text{ wins 3 out of 4}) \\ &= \binom{4}{3} p^3 q p \end{aligned}$$

(This value can also be found from the PMF of a distribution known as the *Negative Binomial*, which we will see later in the course.)

An neater solution is to use the fact (explained in (b)) that we can assume that the teams play all 7 games no matter what. Then let X be the number of wins for team A , so that

$$X \sim \text{Binomial}(7, p).$$

The probability that team A wins the series is

$$P(X \geq 4) = P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7)$$

The PMF of the $\text{Bin}(n, p)$ distribution is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k},$$

and therefore

$$P(X \geq 4) = \binom{7}{4} p^4 q^3 + \binom{7}{5} p^5 q^2 + \binom{7}{6} p^6 q + p^7,$$

which looks different from the above but is actually identical as a function of p (as can be verified by simplifying both expressions as polynomials in p).

(b) Give a clear intuitive explanation of whether the answer to (a) depends on whether the teams always play 7 games (and whoever wins the majority wins the series), or the teams stop playing more games as soon as one team has won 4 games (as is actually the case in practice: once the match is decided, the two teams do not keep playing more games).

The answer to (a) does not depend on whether the teams play all seven games no matter what. Imagine telling the players to continue playing the games even after the match has been decided, just for fun: the outcome of the match won't be affected by this, and this also means that the probability that A wins the match won't be affected by assuming that the teams always play 7 games!

2. A sequence of n independent experiments is performed. Each experiment is a success with probability p and a failure with probability $q = 1 - p$. Show that conditional on the number of successes, all possibilities for the list of outcomes of the experiment are equally likely (of course, we only consider lists of outcomes where the number of successes is consistent with the information being conditioned on).

Let X_j be 1 if the j th experiment is a success and 0 otherwise, and let $X = X_1 + \cdots + X_n$ be the total number of successes. Then for any k and any

$a_1, \dots, a_n \in \{0, 1\}$ with $a_1 + \dots + a_n = k$,

$$\begin{aligned} P(X_1 = a_1, \dots, X_n = a_n | X = k) &= \frac{P(X_1 = a_1, \dots, X_n = a_n, X = k)}{P(X = k)} \\ &= \frac{P(X_1 = a_1, \dots, X_n = a_n)}{P(X = k)} \\ &= \frac{p^k q^{n-k}}{\binom{n}{k} p^k q^{n-k}} \\ &= \frac{1}{\binom{n}{k}}. \end{aligned}$$

This does not depend on $a_1, \dots, a_n$. Thus, for n independent Bernoulli trials, given that there are exactly k successes, the $\binom{n}{k}$ possible sequences consisting of k successes and $n - k$ failures are equally likely. Interestingly, the conditional probability above also does not depend on p (this leads to the notion of a *sufficient statistic*, which is studied in Stat 111).

3. Let $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$, independent of X .

(a) Show that $X + Y \sim \text{Bin}(n + m, p)$, using a story proof.

Interpret X as the number of successes in n independent Bernoulli trials and Y as the number of successes in m more independent Bernoulli trials, where each trial has probability p of success. Then $X + Y$ is the number of successes in the $n + m$ trials, so $X + Y \sim \text{Bin}(n + m, p)$.

(b) Show that $X - Y$ is *not* Binomial.

A Binomial can't be negative, but $X - Y$ is negative with positive probability.

(c) Find $P(X = k | X + Y = j)$. How does this relate to the elk problem from HW 1?

By definition of conditional probability,

$$P(X = k | X + Y = j) = \frac{P(X = k, X + Y = j)}{P(X + Y = j)} = \frac{P(X = k)P(Y = j - k)}{P(X + Y = j)}$$

since the event $X = k$ is independent of the event $Y = j - k$. This becomes

$$\frac{\binom{n}{k} p^k (1 - p)^{n-k} \binom{m}{j-k} p^{j-k} (1 - p)^{m-(j-k)}}{\binom{m+n}{j} p^j (1 - p)^{m+n-j}} = \binom{n}{k} \binom{m}{j-k} / \binom{n+m}{j}.$$

Note that the p disappeared! This is exactly the same distribution as in the elk problem (it is called the *Hypergeometric* distribution). To see why, imagine that there are n male elk and m female elk, each of which is tagged with the word “success” with probability p (independently). Suppose we then want to know how many of the male elk are tagged, given that a total of j elk have been tagged. For this, p is no longer relevant, and we can “capture” the male elk and count how many are tagged, analogously to the original elk problem.

6. Players A and B take turns in answering trivia questions, starting with player A answering the first question. Each time A answers a question, she has probability p_1 of getting it right. Each time B plays, he has probability p_2 of getting it right.

(a) If A answers m questions, what is the PMF of the number of questions she gets right?

The r.v. is $\text{Bin}(m, p_1)$, so the PMF is $\binom{m}{k} p_1^k (1 - p_1)^{m-k}$ for $k \in \{0, 1, \dots, m\}$.

(b) If A answers m times and B answers n times, what is the PMF of the total number of questions they get right (you can leave your answer as a sum)? Describe exactly when/whether this is a Binomial distribution.

Let T be the total number of questions they get right. To get a total of k questions right, it must be that A got 0 and B got k , or A got 1 and B got $k - 1$, etc. These are disjoint events so the PMF is

$$P(T = k) = \sum_{j=0}^k \binom{m}{j} p_1^j (1 - p_1)^{m-j} \binom{n}{k-j} p_2^{k-j} (1 - p_2)^{n-(k-j)}$$

for $k \in \{0, 1, \dots, m + n\}$, with the usual convention that $\binom{n}{k}$ is 0 for $k > n$.

This is the $\text{Bin}(m + n, p)$ distribution if $p_1 = p_2 = p$, as shown in class (using the story for the Binomial, or using Vandermonde’s identity). For $p_1 \neq p_2$, it’s not a Binomial distribution, since the trials have different probabilities of success; having some trials with one probability of success and other trials with another probability of success isn’t equivalent to having trials with some “effective” probability of success.

(c) Suppose that the first player to answer correctly wins the game (with no predetermined maximum number of questions that can be asked). Find the probability that A wins the game.

Let $r = P(A \text{ wins})$. Conditioning on the results of the first question for each player, we have

$$r = p_1 + (1 - p_1)p_2 \cdot 0 + (1 - p_1)(1 - p_2)r,$$

which gives $r = \frac{p_1}{1 - (1 - p_1)(1 - p_2)} = \frac{p_1}{p_1 + p_2 - p_1 p_2}$.

7. A message is sent over a noisy channel. The message is a sequence $x_1, x_2, \dots, x_n$ of n bits ($x_i \in \{0, 1\}$). Since the channel is noisy, there is a chance that any bit might be corrupted, resulting in an error (a 0 becomes a 1 or vice versa). Assume that the error events are independent. Let p be the probability that an individual bit has an error ($0 < p < 1/2$). Let $y_1, y_2, \dots, y_n$ be the received message (so $y_i = x_i$ if there is no error in that bit, but $y_i = 1 - x_i$ if there is an error there).

To help detect errors, the n th bit is reserved for a parity check: x_n is defined to be 0 if $x_1 + x_2 + \dots + x_{n-1}$ is even, and 1 if $x_1 + x_2 + \dots + x_{n-1}$ is odd. When the message is received, the recipient checks whether y_n has the same parity as $y_1 + y_2 + \dots + y_{n-1}$. If the parity is wrong, the recipient knows that at least one error occurred; otherwise, the recipient assumes that there were no errors.

(a) For $n = 5, p = 0.1$, what is the probability that the received message has errors which go undetected?

Note that $\sum_{i=1}^n x_i$ is even. If the number of errors is even (and nonzero), the errors will go undetected; otherwise, $\sum_{i=1}^n y_i$ will be odd, so the errors will be detected.

The number of errors is $\text{Bin}(n, p)$, so the probability of undetected errors when $n = 5, p = 0.1$ is

$$\binom{5}{2}p^2(1-p)^3 + \binom{5}{4}p^4(1-p) \approx 0.073.$$

(b) For general n and p , write down an expression (as a sum) for the probability that the received message has errors which go undetected.

By the same reasoning as in (a), the probability of undetected errors is

$$\sum_{k \text{ even}, k \geq 2} \binom{n}{k} p^k (1-p)^{n-k}.$$

(c) Give a simplified expression, not involving a sum of a large number of terms, for the probability that the received message has errors which go undetected.

Hint for (c): Letting

$$a = \sum_{k \text{ even}, k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} \text{ and } b = \sum_{k \text{ odd}, k \geq 1} \binom{n}{k} p^k (1-p)^{n-k},$$

the binomial theorem makes it possible to find simple expressions for $a + b$ and $a - b$, which then makes it possible to obtain a and b .

Let a, b be as in the hint. Then

$$a + b = \sum_{k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} = 1,$$

$$a - b = \sum_{k \geq 0} \binom{n}{k} (-p)^k (1-p)^{n-k} = (1-2p)^n.$$

Solving for a and b gives

$$a = \frac{1 + (1-2p)^n}{2} \text{ and } b = \frac{1 - (1-2p)^n}{2}.$$

$$\sum_{k \text{ even}, k \geq 0} \binom{n}{k} p^k (1-p)^{n-k} = \frac{1 + (1-2p)^n}{2}.$$

Subtracting off the possibility of no errors, we have

$$\sum_{k \text{ even}, k \geq 2} \binom{n}{k} p^k (1-p)^{n-k} = \frac{1 + (1-2p)^n}{2} - (1-p)^n.$$

Miracle check: note that letting $n = 5, p = 0.1$ here gives 0.073, which agrees with (a); letting $p = 0$ gives 0, as it should; and letting $p = 1$ gives 0 for n odd and 1 for n even, which again makes sense.

1. Let X be a r.v. whose possible values are $0, 1, 2, \dots$, with CDF F . In some countries, rather than using a CDF, the convention is to use the function G defined by $G(x) = P(X < x)$ to specify a distribution. Find a way to convert from F to G , i.e., if F is a known function show how to obtain $G(x)$ for all real x .

Write

$$G(x) = P(X \leq x) - P(X = x) = F(x) - P(X = x).$$

If x is not a nonnegative integer, then $P(X = x) = 0$ so $G(x) = F(x)$. For x a nonnegative integer,

$$P(X = x) = F(x) - F(x - 1/2)$$

since the PMF corresponds to the lengths of the jumps in the CDF. (The $1/2$ was chosen for concreteness; we also have $F(x - 1/2) = F(x - a)$ for any $a \in (0, 1]$.)

Thus,

$$G(x) = \begin{cases} F(x) & \text{if } x \notin \{0, 1, 2, \dots\} \\ F(x - 1/2) & \text{if } x \in \{0, 1, 2, \dots\}. \end{cases}$$

More compactly, we can also write $G(x) = \lim_{t \rightarrow x^-} F(t)$, where the $-$ denotes taking a limit from the left (recall that F is right continuous), and $G(x) = F(\lceil x \rceil - 1)$, where $\lceil x \rceil$ is the “ceiling” of x (the smallest integer greater than or equal to x).

2. There are n eggs, each of which hatches a chick with probability p (independently). Each of these chicks survives with probability r , independently. What is the distribution of the number of chicks that hatch? What is the distribution of the number of chicks that survive? (Give the PMFs; also give the names of the distributions and their parameters, if they are distributions we have seen in class.)



Let H be the number of eggs that hatch and X be the number of hatchlings that survive. Think of each egg as a Bernoulli trial, where for H we define “success” to mean hatching, while for X we define “success” to mean surviving. For example, in the picture above, where  denotes an egg that hatches with the chick surviving,  denotes an egg that hatched but whose chick died, and  denotes an egg that didn’t hatch, the events $H = 7$, $X = 5$ occurred. By the story of the Binomial, $H \sim \text{Bin}(n, p)$, with PMF $P(H = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ for $k = 0, 1, \dots, n$.

The eggs independently have probability pr each of hatching a chick that survives. By the story of the Binomial, we have $X \sim \text{Bin}(n, pr)$, with PMF $P(X = k) = \binom{n}{k} (pr)^k (1 - pr)^{n-k}$ for $k = 0, 1, \dots, n$.

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5. A scientist wishes to study whether men or women are more likely to have a certain disease, or whether they are equally likely. A random sample of m women and n men is gathered, and each person is tested for the disease (assume for this problem that the test is completely accurate). The numbers of women and men in the sample who have the disease are X and Y respectively, with $X \sim \text{Bin}(m, p_1)$ and $Y \sim \text{Bin}(n, p_2)$. Here p_1 and p_2 are unknown, and we are interested in testing the “null hypothesis” $p_1 = p_2$.

(a) Consider a 2 by 2 table listing with rows corresponding to disease status and columns corresponding to gender, with each entry the count of how many people have that disease status and gender (so $m + n$ is the sum of all 4 entries). Suppose that it is observed that $X + Y = r$.

The *Fisher exact test* is based on conditioning on both the row and column sums, so m, n, r are all treated as fixed, and then seeing if the observed value of X is “extreme” compared to this conditional distribution. Assuming the null hypothesis, use Bayes’ Rule to find the conditional PMF of X given $X + Y = r$. Is this a distribution we have studied in class? If so, say which (and give its parameters).

First let us build the 2×2 table (conditioning on the totals m, n , and r).

	Women	Men	Total
Disease	x	$r - x$	r
No Disease	$m - x$	$n - r + x$	$m + n - r$
Total	m	n	$m + n$

Next, let us compute $P(X = x | X + Y = r)$. By Bayes’ rule,

$$\begin{aligned} P(X = x | X + Y = r) &= \frac{P(X + Y = r | X = x)P(X = x)}{P(X + Y = r)} \\ &= \frac{P(Y = r - x)P(X = x)}{P(X + Y = r)}. \end{aligned}$$

Assuming the null hypothesis and letting $p = p_1 = p_2$, we have $X \sim \text{Bin}(m, p)$ and $Y \sim \text{Bin}(n, p)$ with X independent of Y , so $X + Y \sim \text{Bin}(n + m, p)$. Thus,

$$\begin{aligned} P(X = x | X + Y = r) &= \frac{\binom{n}{r-x} p^{r-x} (1-p)^{n-r+x} \binom{m}{x} p^x (1-p)^{m-x}}{\binom{m+n}{r} p^r (1-p)^{m+n-r}} \\ &= \frac{\binom{m}{x} \binom{n}{r-x}}{\binom{m+n}{r}}. \end{aligned}$$

So the conditional distribution is *Hypergeometric* with parameters m, n, r .

(b) Give an intuitive explanation for the distribution of (a), explaining how this problem relates to other problems we’ve seen, and why p_1 disappears (magically?) in the distribution found in (a).

This problem has the same structure as the elk (capture-recapture) problem. In the elk problem, we take a sample of elk from a population, where earlier some were tagged, and we want to know the distribution of the number of tagged elk in the

sample. By analogy, think of the women as corresponding to tagged elk, and men as corresponding to untagged elk. Having r people be infected with the disease corresponds to capturing a new sample of r elk the number of women among the r diseased individuals corresponds to the number of tagged elk in the new sample. Under the null hypothesis and given that $X + Y = r$, the set of diseased people is equally likely to be any set of r people.

It makes sense that the conditional distribution of the number of diseased women does not depend on p , since once we know that $X + Y = r$, we can work directly in terms of the fact that we have a population with r diseased and $m + n - r$ undiseased people, without worrying about the value of p that originally generated the population characteristics.